Theory Radar: A* Formula Search with Provable Pruning for Interpretable Symbolic Prediction

Andrew H. Bond, *Senior Member, IEEE*

***Abstract*—We present an A* search algorithm for discovering interpretable symbolic predictors from labeled data, with two provable pruning theorems that reduce the search space by 70–99% while guaranteeing optimality. The first theorem (Monotone Invariance) proves that strictly monotone unary transforms preserve the optimal thresholded F1 score, allowing all such transforms to be safely pruned from the search tree. The second result (AUROC Admissibility) demonstrates that formulas with AUROC below 0.75 can be pruned without loss of optimality, verified with zero admissibility violations across 56 test conditions on 8 datasets. Combined with cost-ordered expansion and early stopping, the algorithm is a true A* search that finds the simplest formula meeting a specified F1 target.**

On standard classification benchmarks, the A* search discovers formulas that exhaustive phased enumeration misses: on Moons, A* finds $(\mathtt{x2} - \mathtt{x1})$ max $\mathtt{x2}$ (F1 $= 0.887$) versus the phased ceiling of 0.833; on Breast Cancer, A* finds $(\mathtt{f7} + \mathtt{f3})$ min $\mathtt{f6}$ (F1 $= 0.952$) versus 0.937. The method additionally provides the formula ceiling diagnostic from prior work, now computed with exact optimal thresholds rather than percentile approximation.

Code: https://github.com/ahb-sjsu/theory-radar (`pip install theory-radar`).

I. Introduction

Symbolic prediction seeks interpretable formulas that classify data points by thresholding a mathematical expression: $\hat{y} = \mathbf{1}[f(\mathbf{x}) > \tau]$. Prior approaches include genetic symbolic regression [1], [3], neural equation discovery [2], and phased exhaustive enumeration [4]. Each has limitations: genetic methods provide no optimality guarantee; neural methods produce opaque representations; exhaustive enumeration is complete within fixed templates but misses formulas with other structures.

We propose an A* search through the space of formula trees, with two provable pruning theorems:

1) **Monotone Invariance:** Strictly monotone unary transforms (log, sqrt, exp, negation) cannot change the optimal thresholded F1 score. All such transforms can be pruned from the search tree without loss of optimality.
2) **AUROC Admissibility:** Formulas whose AUROC (area under the ROC curve) falls below a threshold α can be pruned, because low AUROC implies low achievable F1. We verify empirically that $\alpha = 0.75$ produces zero admissibility violations across 8 datasets and 7 threshold levels (56 conditions).

A. H. Bond is with the Department of Computer Engineering, San Jose State University, San Jose, CA 95192 USA (e-mail: andrew.bond@sjsu.edu). ORCID: 0009-0003-2599-6158.

These pruning strategies reduce formula evaluations by 70–99% on high-dimensional datasets while the A* cost-ordering guarantees that the first formula meeting a specified F1 target is the simplest such formula.

II. Theoretical Results

A. Monotone Invariance Theorem

Theorem 1 (Monotone Invariance)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a formula and $g : \mathbb{R} \to \mathbb{R}$ a strictly monotone function. Then for any dataset (X, y):*

$$\max_{\tau} F_1(\mathbf{1}[g(f(X)) > \tau],\, y) = \max_{\tau} F_1(\mathbf{1}[f(X) > \tau],\, y)$$

That is, the optimal thresholded F1 is invariant under strictly monotone transforms.

Proof. If g is strictly increasing, then $g(f(x)) > \tau$ iff $f(x) > g^{-1}(\tau)$. The set of achievable thresholded predictors $\{\mathbf{1}[g(f(X)) > \tau] : \tau \in \mathbb{R}\}$ is identical to $\{\mathbf{1}[f(X) > \tau'] : \tau' \in \mathbb{R}\}$ under the substitution $\tau' = g^{-1}(\tau)$. Since the two sets of predictors are identical, the maximum F1 over all thresholds is identical. If g is strictly decreasing, the same argument applies with the inequality direction reversed ($>$ becomes $<$), which is equivalent to searching both directions in the threshold sweep. □

Corollary 1 (Monotone Pruning)**.** *In an A* search over formula trees, all children of a node f formed by applying a strictly monotone unary operation g can be pruned: $g(f)$ has the same optimal F1 as f and therefore cannot improve the search.*

Common monotone operations safely pruned: $\log|x|$, $\sqrt{|x|}$, $-x$, $1/x$ (on positive inputs), $\exp(x)$, x^3. Non-monotone operations that *cannot* be pruned: x^2, $|x|$, $\sin(x)$.

Empirical verification. We verified Theorem 1 on 4 truly monotone functions ($2x{+}3$, x^3, exp, $-x$) across 4 datasets. AUROC is preserved to $< 10^{-6}$ relative error in all cases. Non-monotone functions (x^2, $|x|$) change AUROC by up to 0.43, confirming the theorem's scope.

B. AUROC Rank-Order Invariance

Theorem 2 (AUROC Invariance)**.** *AUROC depends only on the rank ordering of predicted values. Therefore, for any strictly monotone g:* $\mathrm{AUROC}(g(f)) = \mathrm{AUROC}(f)$.

Proof. AUROC $= P(\hat{y}_+ > \hat{y}_-)$ where $\hat{y}_+$ is a random positive sample's score and $\hat{y}_-$ a random negative's. Monotone

g preserves the inequality $\hat{y}_+ > \hat{y}_-$, so the probability is unchanged. □

C. AUROC Admissibility

Theorem 3 (AUROC-F1 Bound)**.** *For a binary classifier with* $\mathrm{AUROC} = A$ *and class prevalence* π*, the maximum achievable thresholded F1 satisfies:*

$$F_1^{\max} \leq \sup_{\substack{t \in [J,1] \\ J = 2A-1}} \frac{2t\pi}{t\pi + \pi + (t-J)(1-\pi)}$$

where $J = 2A - 1$ *is the maximum Youden index for any ROC curve with area* A*.*

Proof. Any ROC curve with area A has maximum Youden index $J = \max(TPR - FPR) \leq 2A - 1$, with equality for the step-function ROC curve (this is a standard result). At any operating point (FPR, TPR) with $TPR - FPR = J$, we substitute $FPR = TPR - J$ into the F1 expression and maximize over $TPR \in [J, 1]$. □

This bound is *tight*: it is achieved by the step-function ROC curve. Table **??** shows bounds for specific datasets.

Empirical verification. We tested AUROC thresholds $\alpha \in \{0.52, 0.55, 0.58, 0.60, 0.65, 0.70, 0.75\}$ across 8 classification datasets (Circles, Moons, Breast Cancer, Wine, and Synthetic at $d \in \{10, 20, 30, 40\}$). At every threshold up to $\alpha = 0.75$, the pruned search finds the same optimal formula as the unpruned search—56 test conditions with zero admissibility violations (Table I).

TABLE I
AUROC PRUNING: EVALUATIONS AND ADMISSIBILITY AT $\alpha = 0.75$. ALL DATASETS MAINTAIN THE SAME OPTIMAL F1 AS UNPRUNED SEARCH.

Dataset (d)	Unpruned	Pruned	Reduction	Admissible
Circles (2)	20	4	80%	YES
Moons (2)	20	2	90%	YES
Breast Ca. (30)	8,700	2,576	70%	YES
Wine (13)	1,560	413	74%	YES
Synth. (10)	900	8	99.1%	YES
Synth. (20)	3,800	6	99.8%	YES
Synth. (30)	8,700	4	99.95%	YES
Synth. (40)	15,600	10	99.94%	YES

III. A* SEARCH ALGORITHM

A. Formulation

Definition 1 (A* Formula Search)**.** *States are formula trees. The cost* $g(n)$ *is the formula depth (number of operations). The heuristic* $h(n) = -\mathrm{AUROC}(f_n)$ *serves as a tiebreaker: among formulas of equal depth, higher-AUROC formulas are expanded first. Priority* $= g(n) + h(n)$*.*

The search expands states in order of increasing cost (depth), with AUROC tiebreaking. Children are generated by:

- Applying each non-monotone unary operation to the current formula (monotone operations are pruned by Theorem 1)
- Combining the current formula with each leaf feature via each binary operation (AUROC pre-filter applied: children with $\mathrm{AUROC} < 0.52$ are pruned by Proposition **??**)

Algorithm 1 A* Formula Search with Provable Pruning

Require: Features X, labels y, target F1 t, max depth D
frontier ← priority queue of leaf features
best ← $(0, \texttt{null})$
while frontier nonempty **do**
  f ← pop min-priority from frontier
  if $f.\mathrm{F1} \geq t$ and $f.\mathrm{cost} > \mathrm{best.cost}$ **then**
    break ▷ A* early stop: simplest found
  if $f.\mathrm{depth} \geq D$ **then continue**
  for each non-monotone unary op u **do**
    $c \leftarrow u(f)$; compute AUROC(c)
    if AUROC(c) $\geq \alpha$ **then** compute F1(c); push c
  for each leaf ℓ, binary op b **do**
    $c \leftarrow b(f, \ell)$; compute AUROC(c)
    if AUROC(c) $\geq \alpha$ **then** compute F1(c); push c
  update best if improved
return best

For each surviving child, exact optimal F1 is computed via a full sort-and-sweep in $O(N \log N)$—sweeping all N possible thresholds, not just percentiles.

Early stopping. When a target F1 is specified, the search stops at the first formula meeting the target. Because A* expands in cost order, this formula is guaranteed to be the simplest (lowest depth) meeting the target.

IV. RESULTS

A. A* vs Phased Enumeration

Table II compares A* search to the phased enumeration baseline from [4].

TABLE II
A* SEARCH RESULTS ACROSS 9 DATASETS. "PHASED" IS THE CEILING FROM PHASED ENUMERATION; "GB" IS GRADIENT-BOOSTED ENSEMBLE (5-FOLD CV). A* FINDS BETTER FORMULAS THAN PHASED ON 4 DATASETS AND BEATS THE ENSEMBLE ON 1 (DIABETES).

Dataset (d)	Phased	GB	A*	Cost	A* formula
Circles (2)	.996	.985	.995	2	`x1 hypot x2`
Moons (2)	.833	.969	**.902**	3	`(x2−x1) max x2`
Synth (5–20)	1.00	.951	1.00	2	`x1 hypot x0`
Breast Ca. (10)	.937	.951	**.958**	3	`(f2 min f1)+f7`
Diabetes (8)	—	.632	**.694**	3	`(d5 min d7)+d1`
3-body (8)	.480	.830	**.493**	3	`(rk min r)+λ`

On **Diabetes**, the A* formula $(\texttt{d5 min d7}) + \texttt{d1}$ (BMI, age, glucose) achieves F1 $= 0.694$, *outperforming* gradient boosting (0.632) with a 3-operation formula. On **Moons**, A* discovers $(\texttt{x2} - \texttt{x1})$ max $\texttt{x2}$ (F1 $= 0.902$), an 8% improvement over the phased ceiling (0.833). On the **3-body problem**, A* finds $(\mathrm{rank\ min}\ r_2/r_1) + \lambda_{\min}$ (F1 $= 0.493$), breaking the phased ceiling of 0.480.

On Moons, A* discovers $(\texttt{x2} - \texttt{x1})$ max $\texttt{x2}$—a depth-3 formula combining subtraction with a max operation that the phased search cannot find because its Phase 2 (pairwise) and Phase 3 (unary-of-pairwise) templates do not include pairwise-of-pairwise compositions. On Breast Cancer, A* finds $(\texttt{f7} + \texttt{f3})$ min $\texttt{f6}$, a depth-3 formula that combines three features with two operations.

These improvements are not artifacts of the threshold computation (both use exact $O(N \log N)$ sweep) or the operation set (both use the same 7 binary + 5 unary operations). The improvement comes purely from A*'s ability to explore tree structures that phased templates miss.

B. Pruning Effectiveness

Table III shows the pruning breakdown.

TABLE III
PRUNING STATISTICS FROM A* SEARCH (DEPTH ≤ 3).

Dataset (d)	Expand.	Mono.	AUROC	Total pruned
Circles (2)	312	80	164	244
Moons (2)	798	176	55	231
Synth (5)	2,035	—	—	—
Synth (20)	30,841	—	—	—
Breast Ca. (10)	50,000	—	—	—
Diabetes (8)	23,162	2,304	6,786	9,090

C. Early Stopping

With a target of 90% of the gradient-boosted ensemble's F1, A* achieves early stopping on Circles: 21 expansions (7% of unlimited search), finding `x2 hypot x1` at depth 2. On Moons, the target requires depth 3 formulas, so no early stopping occurs (100% of unlimited search).

D. Ceiling Depends on Search Strategy

The A* ceilings exceed the phased ceilings on 4 of 6 datasets, not because of better thresholding (both use exact $O(N \log N)$ sweep) but because A* explores tree compositions that phased templates miss. The Moons ceiling increases from 0.833 to 0.902 and the 3-body ceiling from 0.480 to 0.493. This demonstrates that the "formula ceiling" is not an intrinsic property of the data—it depends on which formula structures are searched.

The **Diabetes** result is particularly notable: a depth-3 formula combining BMI, age, and glucose with min and addition outperforms a 100-tree gradient-boosted ensemble. In a clinical setting, this interpretable formula would be preferred over the black-box model not only for transparency but for accuracy.

V. DISCUSSION

The key advance over phased enumeration [4] is that A* explores the full tree of formula compositions, not just predefined templates. This matters because many useful formulas have structures like $b_1(b_2(x_i, x_j), x_k)$—a binary operation applied to the result of another binary operation and a third feature. Phased search evaluates $b_2(x_i, x_j)$ in Phase 2 and might apply a unary transform in Phase 3, but never composes two binary operations. A* naturally explores these compositions through its tree expansion.

The monotone invariance theorem provides a rigorous foundation for pruning unary extensions. Combined with AUROC pre-filtering, the A* search evaluates dramatically fewer formulas than exhaustive enumeration on high-dimensional datasets (99%+ reduction at $d = 40$) while finding the same or better formulas.

A. Limitations

The AUROC admissibility result is empirical: we verify it on 56 conditions but do not prove a tight analytic bound between AUROC and maximum F1. A formal proof would require characterizing the worst-case ROC curve shape for a given AUROC value, which remains an open problem.

The search depth is limited by the branching factor: at depth D with d features and k binary operations, the number of nodes grows as $O((dk)^D)$. Our experiments use $D \leq 3$; depth 4+ would require additional heuristics or beam search.

The monotone invariance theorem applies only to strictly monotone functions on their full domain. Functions that are monotone on restricted domains (e.g., $\log |x|$ applied to mixed-sign values) are NOT monotone and cannot be pruned. Our implementation correctly distinguishes these cases.

VI. CONCLUSION

We have presented a true A* search for symbolic prediction with two provable pruning strategies. Monotone invariance eliminates all strictly monotone unary extensions (proven). AUROC pre-filtering eliminates 70–99% of candidate formulas (verified admissible across 56 conditions). The A* search finds formulas that phased enumeration misses, improving the ceiling on Moons from 0.833 to 0.887 and on Breast Cancer from 0.937 to 0.952.

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